

Lodestar: self-service guide

Explore the supply-chain command centre prototype at your own pace · about 15 minutes

What Lodestar is

Lodestar is a working prototype of a supply-chain command centre for medical imaging. It answers one question every morning: **what is happening in the world right now that could stop us building, shipping or installing a system, and what should we decide about it today?**

- **Everything outside our walls is real public data:** shipping transits, export controls, earthquakes, travel advisories, FDA notices. Each item shows its source and date, and most link straight to it.
- **Everything inside our walls is synthetic for now:** stock cover, installs, revenue and plant figures are illustrative, and marked **synth** wherever they appear.
- **The AI reads the evidence and recommends; people decide.** Every AI-written item shows the evidence it came from.

Getting started

1. Click any of the four links below. The first time, you'll be asked for the **password** (sent to you separately); your browser remembers it for 30 days.
2. Each link opens Lodestar already set to one product line and map view. Work through them in order.
3. Click the **Layers** header (bottom left of the map) to fold that panel away and see more of the map.
4. The data refreshes through the day, so the numbers you see may differ from the examples here.

1. All product lines

the big picture

2. MR

helium and Hormuz

3. CT

export controls

4. Ultrasound

a calm day

1. The big picture

~3 minutes · [link 1](#)

Read the map

- The **glow** marks exposure: places where live events meet our supply chain. Only items scoring 35 or more (out of 100) glow; the number on a marker is its score.
- The **lines** are our shipping lanes. Sea lanes follow real routes through the chokepoints they cross; air lanes are arcs. Brighter lines are more exposed.
- **Ports** are labelled, for example Charleston, with lanes to the UK, the Gulf and Brazil.

Try this

- **Hover over the Strait of Hormuz marker** (the brightest glow, in the Gulf). The tooltip shows the latest daily count of ships through the strait, from IMF PortWatch.

What to notice: in 2025 this strait averaged about 85 ships a day. Compare that with the figure in the tooltip.

- **Read the Command brief** (right-hand side). The large figure is revenue at risk over 90 days (synthetic). Below it, the AI summarises the day and proposes decisions, each with an owning function, a decide-by date and options, recommended first.
- **Scan "Hotspots worth diving into":** every item scored, highest first, each with its evidence and source.

Good to know

- **Synthetic vs real:** anything marked **synth** is illustrative. Connect ERP, the install backlog and the supplier master, and those numbers become real.
- **Shipping data is daily**, not live ship tracking; counts are published a couple of days behind.
- **The AI is advisory** and can make mistakes. Every recommendation shows its evidence so you can check it before acting.
- **Telegram posts are unverified** and shown for context only.

2. MR

~5 minutes · link 2

Start with the product board

Find the **Product exposure board**. The conventional-magnet MR scores high; the sealed low-helium magnet scores low.

What to notice: the sealed magnet uses under 1% of the helium, so it barely registers. The hedge is visible in the data.

Open the evidence behind a hotspot

In **Hotspots worth diving into**, click **Strait of Hormuz**, then **Open evidence drawer**. Scroll down through it:

1. **The AI's read:** a headline and three sentences written from the evidence below it.
2. **Chatter:** recent Telegram posts from news and open-source channels. Unverified, shown for context, never used in the score.
3. **Why this was flagged:** each piece of evidence, tagged live, reference or synthetic, with a bar for how strongly it counts. Click an item to open its source.
4. **How it reaches us:** from the chokepoint, to the lanes through it, to helium, to the plants, products and installs.
5. **Survive vs recover:** how long our stock lasts against how long supply takes to recover (synthetic).

Try this: drag the **what-if** slider from 2 months to 3 and watch installs short and revenue at risk climb. The formula is printed underneath.

6. **What we can do:** costed options, recommended first, with a decide-by date and an owner.

Have the agent draft the next step

Under **Have the agent draft it**, click **S&OP escalation**. A draft note appears in a few seconds, marked "**needs your approval · nothing has been sent**". Nothing is ever sent from Lodestar. "Approve & route" is for show only.

3. CT

~4 minutes · link 3

Click ← **Back to map** to close the drawer, then open link 3.

- **Product board:** CT is also exposed to Hormuz, through the Beijing → Gulf shipping lane. Photon-counting CT is driven by tungsten.
- **Input clock:** compare tungsten and gadolinium.

What to notice: both are under China's export controls, but only tungsten is **exposed** (stock runs out before supply recovers). Gadolinium is **covered**. Same kind of risk, different answer, because the stock cover differs.

- **Open the Gadolinium drawer** from the hotspots list. China's October 2025 expansion of these controls is suspended until about **10 November 2026**: a date worth watching.
- **Regulatory & trade watch:** live FDA recalls and clearances, and the export controls on critical inputs, each tagged sourced or estimate.
- On the map, **Critical Minerals** shows Bayan Obo, the world's largest rare-earth mine, north of our gadolinium source at Baotou.

4. A calm day: ultrasound

~1 minute · link 4

Ultrasound scores low, with nothing at risk. The Command brief calls it a calm day and proposes no decisions.

What to notice: the system also tells you what *not* to worry about.

Explore further

- Use the **product buttons** in the top bar (All · MR · CT · MI · US · XR) to switch product line. MI covers PET/CT and SPECT/CT; XR covers mammography and digital X-ray.
- Click any hotspot on the map or in the list to open its evidence.
- Switch map layers on and off in the **Layers** panel.